

An Unsteady, Confined, Beltrami Cyclone in $\mathbb{R}^3$

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Abstract

This paper investigates the emergence and diffusion of secondary flow in an axially symmetric rotating fluid with a no-slip condition at the base and sidewall. We analyze the Beltrami field, $\vec{u}(r, z, t) = \langle -\frac{\alpha}{r} \partial_z \psi, \frac{\psi}{r}, \frac{\alpha}{r} \partial_r \psi \rangle$ as an exact solution to the Navier-Stokes equation in cylindrical coordinates with the intent to simulate the development of convection in the meridional "r - z" plane. This brief derivation of the Beltrami flow model simplifies a characteristic phenomenon known in meandering rivers, channels through a circular bend, and the "Tea-leaf" paradox explored by Einstein in 1926 [2].

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1 Introduction

Suppose that we confine a volume of initially rotating fluid within a cylindrical boundary. As the fluid rotates, it experiences shear stress against the sidewall and torsional stress at the circular base. Depending on the depth, z , the base friction can greatly impact how rapidly each velocity component decays and the degree to which convection originates. The machinery of the rotational vortex solution perceives a similar flow type to that of Bödewaft [3], though for simplicity, we will not explore the Bödewaft boundary layer.

First, we seek a general solution to the laminarized, pressureless Navier-Stokes equation that satisfies these two Dirichlet boundary conditions, and second, we solve two types of initial swirls in Sections (3.1) and (3.2):

1. Irrotational, resembling that of Oseen-Lamb's vortex, and,
2. Rotational, resembling a rigid-body, centrifuged flow.

And third, we find an approximate solution to the scalar function, $\alpha(t)$, that ensures the meridional velocities are initially zero by $\alpha(0) = 0$.

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2 The Beltrami Flow Model

Consider a velocity field, $\vec{u} = \langle u_r(r, z, t), u_\theta(r, z, t), u_z(r, z, t) \rangle$ in cylindrical coordinates. Negating pressure and gravity, and invoking axisymmetry ($\partial_\theta \vec{u} = 0$), the azimuthal momentum equation is [6],

$$\begin{aligned} \frac{\partial u_\theta}{\partial t} + \vec{u} \cdot \nabla u_\theta &= \nu \left(\nabla^2 u_\theta - \frac{u_\theta}{r^2} \right), \quad \nabla \cdot \vec{u} = 0 \\ \equiv \frac{\partial u_\theta}{\partial t} + u_r \frac{\partial u_\theta}{\partial r} + u_z \frac{\partial u_\theta}{\partial z} + \frac{u_r u_\theta}{r} &= \nu \left(\frac{\partial^2 u_\theta}{\partial r^2} + \frac{1}{r} \frac{\partial u_\theta}{\partial r} + \frac{\partial^2 u_\theta}{\partial z^2} - \frac{u_\theta}{r^2} \right) \end{aligned} \quad (1)$$

A solution to (1) is given by a Beltrami flow, meaning the vorticity and velocity fields are parallel and proportioned by a scalar function, $\alpha(\mathbf{x}, t)$.

$$\begin{aligned} \vec{\omega} &= \nabla \times \vec{u} = \alpha \vec{u} \\ \equiv \begin{bmatrix} \omega_r \\ \omega_\theta \\ \omega_z \end{bmatrix} &= \begin{bmatrix} -\frac{\partial u_\theta}{\partial z} \\ \frac{\partial u_r}{\partial z} - \frac{\partial u_z}{\partial r} \\ \frac{1}{r} \frac{\partial(r u_\theta)}{\partial r} \end{bmatrix} = \alpha \begin{bmatrix} u_r \\ u_\theta \\ u_z \end{bmatrix} \end{aligned} \quad (2)$$

This likewise asserts a zero Lamb vector: $\vec{u} \times \vec{\omega} = 0$. In the case of steady state inviscid flows, one would solve the stream function, $\psi(r, z)$, in the Bragg-Hawthorne equation for a Beltrami field, $\vec{u} = \langle -\frac{1}{r} \partial_z \psi, u_\theta, \frac{1}{r} \partial_r \psi \rangle$. Although it can be useful to isolate u_θ from u_r and u_z , we seek their dependence on u_θ since the azimuthal friction with the floor changes the fluid's angular momentum, hence, generating convection [2]. As is customary with the Bragg-Hawthorne equation, let $\Gamma(\psi) = r u_\theta = \psi$, and let $r u_r = -\alpha(t) \partial_z \psi$ and $r u_z = \alpha(t) \partial_r \psi$ for $\psi(r, z, t)$. Let $\alpha(t)$ be the convection growth coefficient such that $\alpha(0) = 0$. Substituting each component into (1) yields the form [6],

$$\frac{\partial \psi}{\partial t} = \nu \left(\frac{\partial^2 \psi}{\partial r^2} - \frac{1}{r} \frac{\partial \psi}{\partial r} + \frac{\partial^2 \psi}{\partial z^2} \right) \quad (3)$$

which allows the cancellation of $\alpha(t)$.

2.1 Boundary Conditions:

We obtain a solution to (3) by the following Dirichlet (no-slip) boundary conditions. We can separate the variables as $\psi(r, z, t) = R(r, t)T(z, t)$, where,

$$\begin{cases} u_\theta(0, z, t) = 0, & u_\theta(R_f, z, t) = 0, \quad R_f \in \mathbb{R}^+ \\ u_\theta(r, z, 0) = \frac{1}{r} R(r, 0)T(z, 0) \neq 0, & u_\theta(r, 0, t) = 0, \quad 0 < R_f \end{cases}$$

Thus, for $R(r, t)T(z, t) \in u_\theta(r, z, t)$,

$$\left\{ \begin{aligned} \lim_{r \rightarrow 0} \frac{R(r, t)}{r} &= 0, & T(z, 0) &= \lim_{z \rightarrow \infty} T(z, t) = 1 \\ R(R_f, t) &= 0, & T(0, t) &= \lim_{t \rightarrow \infty} T(z, t) = 0 \\ \frac{R(r, 0)}{r} &= f(r), & f(r) &= u_\theta(r) \quad (= \frac{\Gamma_0}{2\pi r} \text{ or } 2\omega_0) \end{aligned} \right\} \quad (4)$$

where Γ_0 is the irrotational vortex's circulation, ω_0 is the rotational vortex's angular velocity (twice its vorticity), and R_f is the sidewall radius.

3 The Stream Function Solution:

Substituting $\psi(r, z, t) = R(r, t)T(z, t)$ into Equation (3),

$$\begin{aligned} \frac{\partial (R(r, t)T(z, t))}{\partial t} &= \nu \left(\frac{\partial^2 (R(r, t)T(z, t))}{\partial r^2} - \frac{1}{r} \frac{\partial (R(r, t)T(z, t))}{\partial r} + \frac{\partial^2 (R(r, t)T(z, t))}{\partial z^2} \right) \\ \Leftrightarrow T \frac{\partial R}{\partial t} + R \frac{\partial T}{\partial t} &= \nu \left(T \frac{\partial^2 R}{\partial r^2} - \frac{T}{r} \frac{\partial R}{\partial r} + R \frac{\partial^2 T}{\partial z^2} \right) \\ \Leftrightarrow R \frac{\partial T}{\partial t} - \nu R \frac{\partial^2 T}{\partial z^2} &= \nu T \frac{\partial^2 R}{\partial r^2} - \nu \frac{T}{r} \frac{\partial R}{\partial r} - T \frac{\partial R}{\partial t} \\ \Leftrightarrow \frac{1}{T} \frac{\partial T}{\partial t} - \nu \frac{1}{T} \frac{\partial^2 T}{\partial z^2} &= \frac{\nu}{R} \frac{\partial^2 R}{\partial r^2} - \frac{\nu}{rR} \frac{\partial R}{\partial r} - \frac{1}{R} \frac{\partial R}{\partial t} \end{aligned}$$

we observe that both sides of the PDE are functionally disjoint. Suppose that they are equal to some constant, $k \in \mathbb{R}$. Then,

$$\begin{aligned}\frac{\partial T}{\partial t} - \nu \frac{\partial^2 T}{\partial z^2} &= kT(z, t) \\ \nu \left(\frac{\partial^2 R}{\partial r^2} - \frac{1}{r} \frac{\partial R}{\partial r} \right) - \frac{\partial R}{\partial t} &= kR(r, t)\end{aligned}$$

If $k = 0$, then $R(r, t)$ and $T(z, t)$ are solutions to simpler one-dimensional linear diffusion equations [5]:

$$\begin{aligned}\frac{\partial T}{\partial t} &= \nu \frac{\partial^2 T}{\partial z^2} \\ \frac{\partial R}{\partial t} &= \nu \left(\frac{\partial^2 R}{\partial r^2} - \frac{1}{r} \frac{\partial R}{\partial r} \right)\end{aligned}\tag{5}$$

Solving for $T(z, t)$, suppose we use the similarity solution, $T(r, t) = f(\eta)$, where $\eta = \frac{z}{\sqrt{\nu t}}$, as derived by dimensional analysis,

$$\frac{T}{t} \sim \nu \frac{T}{z^2} \Rightarrow z \sim \sqrt{\nu t}$$

Taking each derivative of $f(\eta)$,

$$\begin{aligned}\frac{\partial T}{\partial t} &= \frac{\partial f}{\partial \eta} \frac{\partial \eta}{\partial t} = -f' \frac{zt^{-3/2}}{2\sqrt{\nu}} = -f' \frac{\eta}{2t} \\ \frac{\partial^2 T}{\partial z^2} &= \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial \eta} \frac{\partial \eta}{\partial z} \right) \\ &= \frac{\partial f}{\partial \eta} \underbrace{\frac{\partial}{\partial z} \left(\frac{\partial \eta}{\partial z} \right)}_{=0} + \frac{\partial \eta}{\partial z} \frac{\partial}{\partial z} \left(\frac{\partial f}{\partial \eta} \right) \\ &= \frac{\partial \eta}{\partial z} \cdot \frac{\partial^2 f}{\partial \eta^2} \frac{\partial \eta}{\partial z} \\ &= f'' \frac{1}{\nu t}\end{aligned}\tag{6}$$

the PDE for $T(z, t)$ by substitution of (6) into (5) becomes,

$$\begin{aligned}-f' \frac{\eta}{2t} &= \nu f'' \frac{1}{\nu t} \\ f'' &= -f' \frac{\eta}{2} \quad , \quad \text{let, } u(\eta) = f'(\eta) \\ \frac{du}{d\eta} &= -u \frac{\eta}{2} \\ \int \frac{du}{u} &= -\frac{1}{2} \int \eta d\eta \\ u(\eta) &= e^{-\frac{1}{4}\eta^2 + C} \\ \int df(\eta) &= C_1 \int e^{-\frac{1}{4}\eta^2} d\eta \\ f(\eta) &= C_1 \sqrt{\pi} \operatorname{erf}(\eta) + C_2 \\ T(z, t) &= C_1 \sqrt{\pi} \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) + C_2\end{aligned}$$

Using the boundary condition (4), $\lim_{z \rightarrow \infty} T(z, t) = 1$, thus $C_1 = 1/\sqrt{\pi}$ and $C_2 = 0$. Hence, we obtain,

$$T(z, t) = \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right)\tag{7}$$

Solving for $R(r, t)$, we can use separation of variables again, letting $R(r, t) = u(r)v(t)$. Our boundary conditions from (4) yield,

$$\begin{cases} R(r, 0) = u(r), & u(0) = 0 \\ R(0, t) = 0, & v(0) = 1 \end{cases}$$

Substituting this into (5), we have,

$$\begin{aligned} \frac{d(u(r)v(t))}{dt} &= \nu \left(\frac{d^2(u(r)v(t))}{dr^2} - \frac{1}{r} \frac{d(u(r)v(t))}{dr} \right) \\ \Leftrightarrow u(r)v'(t) &= \nu \left(v(t)u''(r) - \frac{1}{r}v(t)u'(r) \right) \\ \Leftrightarrow \frac{v'(t)}{\nu v(t)} &= \frac{u''(r)}{u(r)} - \frac{u'(r)}{ru(r)} \end{aligned}$$

Again, we see that both sides of the differential equation are disjoint, and thus each is equal to some constant, $K \in \mathbb{R}$. Then the left-hand side becomes

$$\begin{aligned} \frac{v'(t)}{\nu v(t)} &= K \\ \int \frac{dv}{v} &= \int K \nu dt \\ \ln |v| &= K \nu t + C \\ \therefore v(t) &= e^{K \nu t}, \quad v(0) = 1 = e^C \end{aligned}$$

and the right-hand side is a Bessel differential equation, which has the solution in terms of first-order Bessel functions of the first and second kind.

$$\begin{aligned} K &= \frac{u''(r)}{u(r)} - \frac{u'(r)}{ru(r)} \\ \Leftrightarrow 0 &= ru''(r) - u'(r) - Kru(r) \\ \therefore u(r) &= Ar J_1(\sqrt{-K}r) + Br Y_1(\sqrt{-K}r) \end{aligned}$$

Since $\lim_{r \rightarrow 0} Y_1(\sqrt{-K}r)$ is undefined for all $K \in \mathbb{R}$, $B = 0$. Furthermore, let $K = -\bar{\lambda}^2$ if we consider only the real solutions to (3). Then $R(r, t)$ becomes a general solution,

$$R(r, t) = Cr J_1(\bar{\lambda}r) e^{-\bar{\lambda}^2 \nu t}$$

Using superposition, we can obtain all possible solutions to (3) in the form of a Fourier-Bessel series.

$$R(r, t) = \sum_{k=1}^{\infty} C_k r J_1(\bar{\lambda}_k r) e^{-\bar{\lambda}_k^2 \nu t}$$

Given the boundary condition, $u_\theta(R_f, z, t) = 0$ in (4), $\bar{\lambda}_k$ are all roots of the Bessel function such that

$$\begin{aligned} J_1(\bar{\lambda}_k r) &= 0, \quad r \in [0, 1] \\ \bar{\lambda}_k &= \frac{\lambda_k}{R_f} \Rightarrow J_1\left(\frac{\lambda_k}{R_f} r\right) = 0, \quad r \in [0, R_f] \end{aligned}$$

as computed in the Bessel function tables [1].

Table 1: *Table of the first 15 roots of $J_1(x)$, where $\lambda_k \approx 3.14533k + 0.730956$.*

$\lambda_1 = 3.831706$	$\lambda_6 = 19.615859$	$\lambda_{11} = 35.33231$
$\lambda_2 = 7.015587$	$\lambda_7 = 22.760084$	$\lambda_{12} = 38.47477$
$\lambda_3 = 10.173468$	$\lambda_8 = 25.903672$	$\lambda_{13} = 41.61709$
$\lambda_4 = 13.323692$	$\lambda_9 = 29.046829$	$\lambda_{14} = 44.75932$
$\lambda_5 = 16.470630$	$\lambda_{10} = 32.189680$	$\lambda_{15} = 47.90146$

Thus far, with Equation (7), the streamfunction is,

$$\psi(r, z, t) = \text{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) \sum_{k=1}^{\infty} C_k r J_1\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \quad (8)$$

In Sturm-Liouville theory [7], because u_θ is an eigenfunction expansion, $J_1\left(\frac{\lambda_k}{R_f} r\right)$ forms a basis of eigenfunctions in the Fourier Bessel series when $z \rightarrow \infty$ and $t \rightarrow 0$. Since in (4), $\frac{R(r,0)}{r} = f(r)$,

$$f(r) = \sum_{k=1}^{\infty} C_k J_1\left(\frac{\lambda_k}{R_f} r\right) \quad (9)$$

by the orthogonality relation [4] on a function, $f(x)$, for $x \in [0, \alpha]$. Proceeding from the orthonormal relation, the Fourier-Bessel coefficient is:

$$C_m = \frac{2}{\alpha^2 J_{\nu+1}^2(\lambda_m)} \int_0^\alpha x f(x) J_\nu\left(\frac{\lambda_m}{\alpha} x\right) dx, \quad x \in [0, \alpha] \quad (10)$$

We find these coefficients in Equation (9) by considering the two types of initial cyclonic flows: the rotational and irrotational vortices.

3.1 The Initially Rotational Vortex

For all $z > 0$, we let the constant rotational vortex be our initial condition.¹

$$u_\theta(r, 0) = f(r) = \frac{\Gamma_0}{2\pi R_f^2} r = \omega_0 r \quad (11)$$

Γ_0 is the circulation and ω_0 is the initial angular velocity of the rotational flow. Letting $f(x)$ be Equation (11), $\alpha = R_f$, and $\lambda_m = \lambda_k$, we find from using (10) to solve for C_k in (9) [8],

$$\begin{aligned} C_k &= \frac{2}{R_f^2 J_2^2(\lambda_k)} \int_0^{R_f} r (\omega_0 r) J_1\left(\frac{\lambda_k}{R_f} r\right) dr, \quad r \in [0, R_f] \\ &= \frac{2\omega_0}{R_f^2 J_2^2(\lambda_k)} \int_0^{R_f} r^2 J_1\left(\frac{\lambda_k}{R_f} r\right) dr \\ &= \frac{2\omega_0}{R_f^2 J_2^2(\lambda_k)} \frac{R_f^3}{\lambda_k^3} \int_0^{\lambda_k} u^2 J_1(u) du, \quad u = \frac{\lambda_k}{R_f} r \\ &= \frac{2\omega_0}{R_f^2 J_2^2(\lambda_k)} \frac{R_f^3}{\lambda_k^3} [u^2 J_2(u)]_0^{\lambda_k} \\ &= \frac{2\omega_0}{R_f^2 J_2^2(\lambda_k)} \frac{R_f^3}{\lambda_k^3} \lambda_k^2 J_2(\lambda_k) \\ &= \frac{2\omega_0 R_f}{\lambda_k J_2(\lambda_k)} \end{aligned}$$

¹Note that $u_\theta(r, 0) = u_\theta(r, z, 0)$ since $\lim_{t \rightarrow 0} \text{erf}\left(\frac{z}{\sqrt{\nu t}}\right) = 1$, thus $\forall z > 0$, the initial condition does not vary with z .

The relation, $\frac{\Gamma_0}{2\pi R_f^2} = \omega_0$, comes from the vorticity flux, $\Gamma = \int_A \vec{\omega}_z \cdot d\mathbf{A} = \int_0^{R_f} \omega_z \cdot 2\pi r dr \sim 2\omega_0 \pi R_f^2$, where $\|\omega_z\| \sim 2\omega_0$. In the rotational vortex, $\|\omega_z\| = 2\omega_0$, and thus, $\Gamma_0 = 2\omega_0 \pi R_f^2$.

the coefficients to be substituted into Equation (8). Hence, the solution is obtained.

$$u_\theta(r, z, t) = 2\omega_0 R_f \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) \sum_{k=1}^{\infty} \frac{J_1\left(\frac{\lambda_k}{R_f} r\right)}{\lambda_k J_2(\lambda_k)} e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \quad (12)$$

3.2 The Initially Irrotational Vortex

Likewise, let,

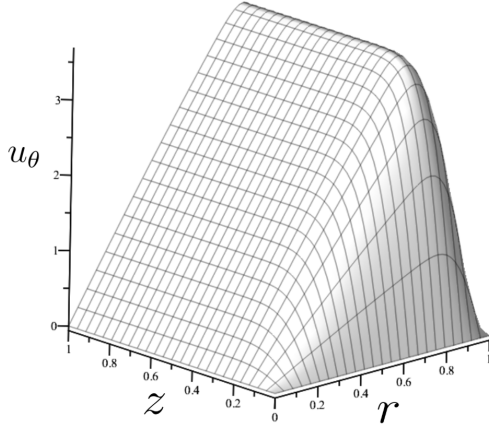
$$u_\theta(r, 0) = f(r) = \frac{\Gamma_0}{2\pi r} \quad (13)$$

be our initial condition. Using (10) again for (13), we obtain

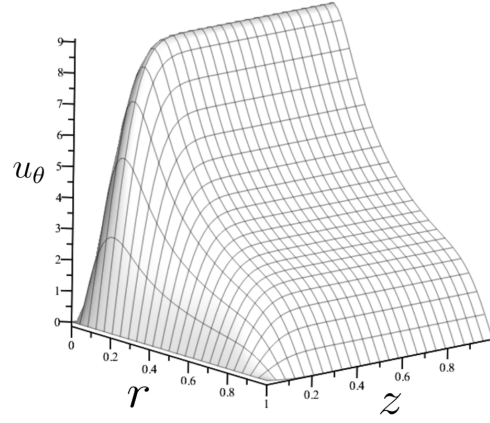
$$\begin{aligned} C_k &= \frac{2}{R_f^2 J_2^2(\lambda_k)} \int_0^{R_f} r \left(\frac{\Gamma_0}{2\pi r} \right) J_1\left(\frac{\lambda_k}{R_f} r\right) dr, \quad r \in [0, R_f] \\ &= \frac{2}{R_f^2 J_2^2(\lambda_k)} \frac{\Gamma_0}{2\pi} \int_0^{R_f} J_1\left(\frac{\lambda_k}{R_f} r\right) dr \\ &= \frac{\Gamma_0}{\pi} \frac{1}{R_f^2 J_2^2(\lambda_k)} \frac{R_f}{\lambda_k} \left[1 - J_0\left(\frac{\lambda_k}{R_f} r\right) \right]_0^{R_f} \\ &= \frac{\Gamma_0}{\pi} \frac{1 - J_0(\lambda_k)}{R_f \lambda_k J_2^2(\lambda_k)} \end{aligned}$$

the coefficients to be substituted into (8). And so,

$$u_\theta(r, z, t) = \frac{\Gamma_0}{\pi} \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) \sum_{k=1}^{\infty} \frac{1 - J_0(\lambda_k)}{R_f \lambda_k J_2^2(\lambda_k)} J_1\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \quad (14)$$



(a) Equation (12) with $\omega_0 = 5 \text{ rad/s}$.



(b) Equation (14) with $\Gamma_0/\pi = 4 \text{ cm}^2/\text{s}$.

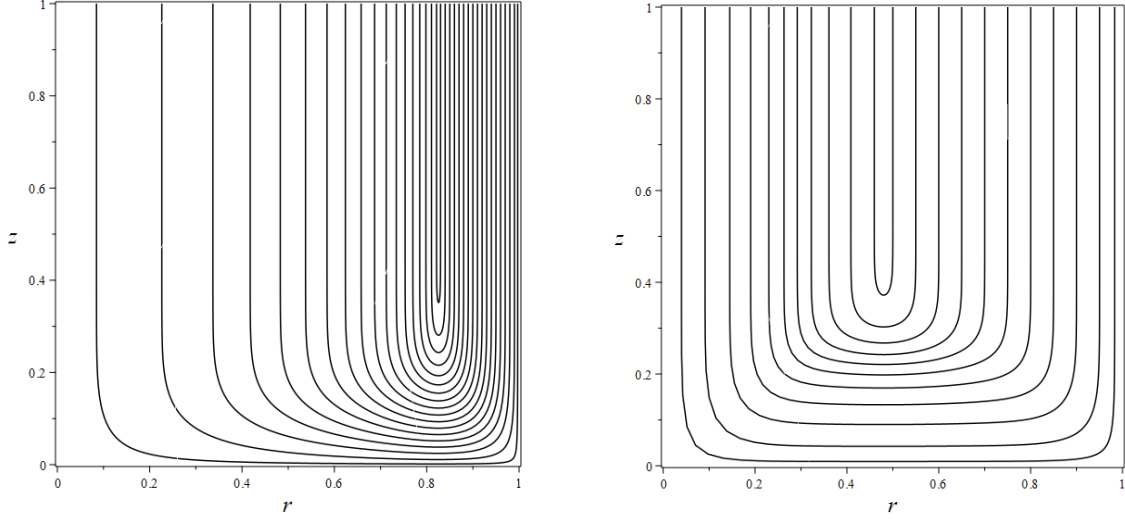
Figure 1: The initially rotational (1a) and irrotational (1b) vortexes at $t = 0.5 \text{ s}$, $\nu = 0.01 \text{ cm}^2/\text{s}$, and $R_f = 1 \text{ cm}$.

4 The Meridional Flow Field

Solving $u_r = -\frac{\alpha(t)}{r} \frac{\partial \psi}{\partial z}$ and $u_z = \frac{\alpha(t)}{r} \frac{\partial \psi}{\partial r}$, we have,

$$u_r(r, z, t) = -\frac{\alpha(t)}{\sqrt{\pi \nu t}} \exp\left\{-\frac{z^2}{4\nu t}\right\} \sum_{k=1}^{\infty} C_k J_1\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \quad (15)$$

$$u_z(r, z, t) = \alpha(t) \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) \sum_{k=1}^{\infty} C_k \frac{\lambda_k}{R_f} J_0\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \quad (16)$$



(a) The rotational vortex with $\omega_0 = 5 \text{ rad/s}$.

(b) The irrotational vortex with $\Gamma_0/\pi = 4 \text{ cm}^2/\text{s}$.

Figure 2: Streamlines generated by Equations (15) and (16) at $t = 0.5 \text{ s}$, $\nu = 0.01 \text{ cm}^2/\text{s}$, $\alpha(t) = 1$, and $R_f = 1 \text{ cm}$. (3a) has circulation centralized in the outer region of the vortex, $r \approx 0.8$, while (3b) has circulation at $r \approx 0.5 \text{ cm}$.

4.1 Surface layer interaction at $z = h$

Suppose we seek a no-slip condition on u_r and u_z at the fluid's surface, $z = h$, for some real constant, $h > 0$. In retrospect, this would ideally force $u_\theta(r, h, t) = 0$. The vertical injection-ejection meridional flow is equivalent to a closed circulation due to mass conservation, and thus, there is little impact on u_θ with an azimuthal slip-condition at $z = h$. In Equation (5), decompose $T(z, t)$ into the sum two functions, $T_1(z, t)$ and $T_2(z - h, t)$, and assign the conditions,

$$\left\{ \begin{array}{l} T(z, 0) = 1, \quad T(h, t) = 0 \\ T(0, t) = 0, \quad \lim_{t \rightarrow \infty} T(z, t) = 0 \end{array} \right\} \quad (17)$$

Letting T_1 and T_2 describe floor and ceiling boundary layers, we have two heat equations

$$\begin{aligned} \frac{\partial T_1}{\partial t} + \frac{\partial T_2}{\partial t} &= \nu \frac{\partial^2 T_1}{\partial z^2} + \nu \frac{\partial^2 T_2}{\partial z^2} \\ \Leftrightarrow \frac{\partial T_1}{\partial t} - \nu \frac{\partial^2 T_1}{\partial z^2} &= -\frac{\partial T_2}{\partial t} + \nu \frac{\partial^2 T_2}{\partial z^2} \end{aligned}$$

with the separated solutions,

$$T_1(z, t) = \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) \quad \text{and,} \quad T_2(z - h, t) = C_1 \sqrt{\pi} \operatorname{erf}\left(\frac{z - h}{2\sqrt{\nu t}}\right) + C_2$$

Applying the boundary conditions in (17), we have,

$$T(z, t) = \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) - \left[\operatorname{erf}\left(\frac{z-h}{2\sqrt{\nu t}}\right) + \operatorname{erf}\left(\frac{h}{2\sqrt{\nu t}}\right)\right] \quad (18)$$

Applying Equation (18) to u_r and u_z , the velocity field may be written as,

$$\vec{u} = \begin{bmatrix} u_r \\ u_\theta \\ u_z \end{bmatrix} = \begin{bmatrix} -\frac{\alpha(t)}{\sqrt{\pi\nu t}} \left(\exp\left\{-\frac{z^2}{4\nu t}\right\} - \exp\left\{-\frac{(z-h)^2}{4\nu t}\right\} \right) \sum_{k=1}^{\infty} C_k J_1\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \\ \operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) \sum_{k=1}^{\infty} C_k J_1\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \\ \alpha(t) \left[\operatorname{erf}\left(\frac{z}{2\sqrt{\nu t}}\right) - \operatorname{erf}\left(\frac{z-h}{2\sqrt{\nu t}}\right) - \operatorname{erf}\left(\frac{h}{2\sqrt{\nu t}}\right) \right] \sum_{k=1}^{\infty} C_k \frac{\lambda_k}{R_f} J_0\left(\frac{\lambda_k}{R_f} r\right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \end{bmatrix} \quad (19)$$

This constructs a symmetric pseudo-velocity field that obeys mass conservation, though it does not conclusively satisfy Equation (1) unless Equation (18) applies to u_θ in (19). The following section will consider Equations (15) and (16) in finding the secondary flow growth coefficient, $\alpha(t)$.

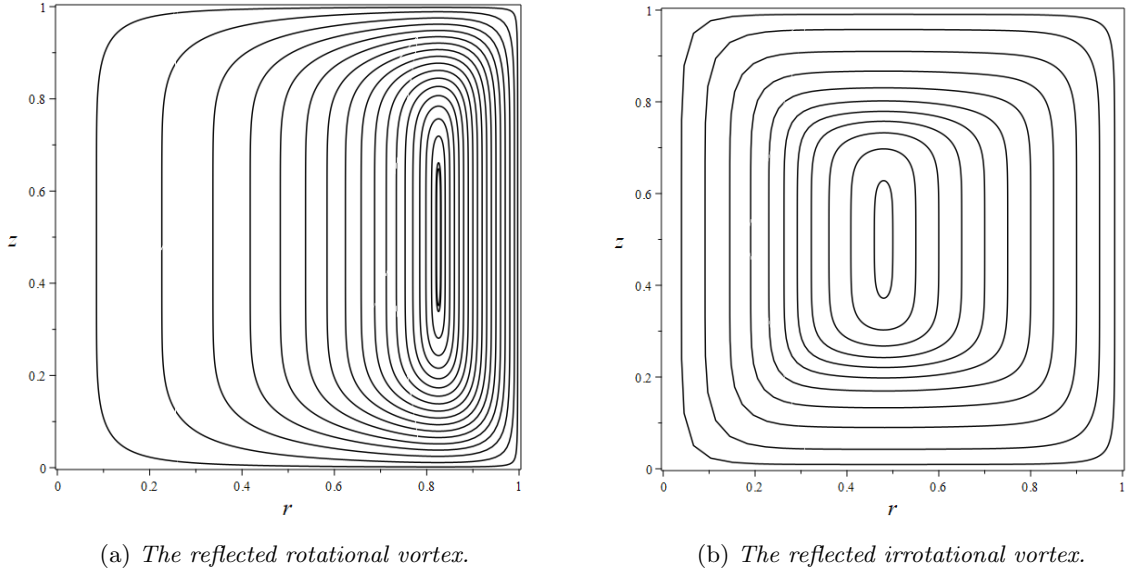


Figure 3: Streamlines generated by Equation (19). (3a) and (3b) share the same features and conditions as in Figure 2.

4.2 Finding An Approximate $\alpha(t)$

The difficulty in finding $\alpha(t)$ begins with substituting Equations (15) and (16) into either the $\hat{r}$ or $\hat{z}$ components of the Navier-Stokes equation. Arbitrarily, let us assign the boundary conditions,

$$\left\{ \begin{array}{l} \alpha(0) = 0 \\ \frac{\partial \alpha}{\partial t} \Big|_{t=0} = \text{const.} \end{array} \right\} \quad (20)$$

Suppose that $\alpha(t)$ is linear, such that,

$$\alpha(t) = kt \quad , \quad k \in \mathbb{R}^+$$

and we impose the conditions

$$|u_r| < |u_\theta| \quad \text{and,} \quad |u_z| < |u_\theta| \quad , \quad \forall r, z, t \in \mathbb{R}$$

because, for all time, the meridional velocities cannot exceed the azimuthal velocity. To ensure these conditions are satisfied, we compare the extremums of u_r and u_z to u_θ . There exists some function, $q(t) \in u_\theta$ as $z \rightarrow \infty$, such that $u_\theta(q(t), z \rightarrow \infty, t)$ is the global maximum of $u_\theta(r, z, t)$, defined as

$$u_{max}(t) := u_\theta(q(t), z \rightarrow \infty, t) = \sum_{k=1}^{\infty} C_k J_1 \left(\frac{\lambda_k}{R_f} q(t) \right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t}, \quad \begin{cases} q(0) = q_0, & q_0 \in [0, R_f] \\ q(t \rightarrow \infty) = q_f, & q_0 \in (0, R_f) \end{cases}$$

For $|u_r| < u_{max}(t)$, $\exists s(t) \in u_r$ such that $u_r(s(t), 0, t)$ is the global maximum of u_r . Then,

$$\begin{aligned} |u_r(s(t), 0, t)| &< u_{max}(t) \\ \Leftrightarrow \frac{\alpha(t)}{\sqrt{\pi \nu t}} \sum_{k=1}^{\infty} C_k J_1 \left(\frac{\lambda_k}{R_f} s(t) \right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} &< \sum_{k=1}^{\infty} C_k J_1 \left(\frac{\lambda_k}{R_f} q(t) \right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \\ \Leftrightarrow s(t) &= q(t), \text{ thus,} \\ \therefore \alpha(t) &< \sqrt{\pi \nu t} \end{aligned}$$

which implies that $k < \sqrt{\frac{\pi \nu}{t}}$. A trivial case is letting $\alpha(t) = \sqrt{\nu t}$, however, by (17), $\alpha'(t \rightarrow 0) < \infty$.

For $|u_z| < u_{max}(t)$, the global maximum is either $u_z(0, z \rightarrow \infty, t)$ or $u_z(R_f, z \rightarrow \infty, t)$. And so,

$$\begin{aligned} &\left. \begin{aligned} &|u_z(0, z \rightarrow \infty, t)| \\ &\text{and,} \\ &|u_z(R_f, z \rightarrow \infty, t)| \end{aligned} \right\} < u_{max}(t) \\ \Leftrightarrow &\left. \begin{aligned} &\alpha(t) \sum_{k=1}^{\infty} C_k \frac{\lambda_k}{R_f} e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \\ &\text{and,} \\ &\alpha(t) \sum_{k=1}^{\infty} C_k \frac{\lambda_k}{R_f} J_0(\lambda_k) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \end{aligned} \right\} < \sum_{k=1}^{\infty} C_k J_1 \left(\frac{\lambda_k}{R_f} q(t) \right) e^{-\frac{\lambda_k^2}{R_f^2} \nu t} \end{aligned}$$

In the rotational vortex, $u_z(0, z \rightarrow \infty, t) < u_{max}(t)$ is trivial because $(q_0 = R_f) > 0$ and thus $J_1 \left(\frac{\lambda_k}{R_f} q_0 \right) > 0$, which is greater than $\alpha(0)$. Additionally, since $u_r(q(t), 0, t) \sim u_{max}(t)$, $u_z(0, z \rightarrow \infty, t) < u_r(q(t), 0, t)$. For brevity and the supposition of $\alpha(t)$, we will not attempt to prove all cases for u_z . Fulfilling the condition in (20), let $k = \sqrt{\frac{\pi \nu}{t + \delta}}$, such that,

$$\alpha(t) = \sqrt{\frac{\pi \nu t^2}{t + \delta}}, \quad \begin{aligned} &\delta \in \mathbb{R}^+ \\ &\delta > 0 \end{aligned}$$

and,

$$\left. \frac{\partial \alpha}{\partial t} \right|_{t=0} = \lim_{t \rightarrow 0} \frac{t + 2\delta}{2\pi \nu} \left(\frac{\pi \nu}{t + \delta} \right)^{\frac{3}{2}} = \sqrt{\frac{\pi \nu}{\delta}}$$

This leaves δ as an unknown constant that governs the initial rate (and subsequently the extent) of secondary flow growth. The true behavior of $\alpha(t)$ has yet to be explored, and will likely require numerical methods or series approximations.

5 Conclusions

While this Beltrami flow model simplifies a complex boundary interaction phenomenon in swirl dynamics, it operates under numerous assumptions that would be difficult to idealize in CFD simulations, such as the natural emergence of boundary-layer separation at the cylindrical base and its instability in large time. A few of these assumptions include,

1. The Lamb vector is always zero, such that $\vec{\omega} \times \vec{u} = 0$ (defining Equation (2)).

2. The meridional flow is vertically symmetric about the r -plane, on which Equations (18) and (19) rely.
3. u_r and u_z can freely slip at $r = R_f$, such that $u_r(R_f, z, t) \neq 0$ and $u_z(R_f, z, t) \neq 0$.
4. $\alpha(t)$ is initially linear, is bounded below $\sqrt{\pi\nu t}$, and is largely assumed to work under the condition, (20).
5. Gravitational and pressure forces are negated, which may influence the flow geometry significantly. For example, the cyclone's updraft (generated by u_z) may significantly decrease with respect to increasing z under gravitational resistance.

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